MACROECONOMICS PRELIM, JULY 2024

ANSWER KEY FOR QUESTIONS 1 AND 2 (ECN 200D)

**Question 1**

a) Since getting the budget constraint right here is very important let's write that separately first. The household has an income which it can allocate to either $c$ or $i$. However, as I clearly implied in the hint, for any extra unit that goes into $i$ the household's income (or resources) will go down by $\tau$. Hence, we have

$$c + i = w + rk - \tau i + T,$$

which can be re-written as

$$c + i(1+\tau) = w + rk + T,$$

and since we know that $k' = (1-\delta)k + i$ (or, if you prefer $i = k' - (1-\delta)k$), the budget constraint becomes

$$c = w + [r + (1-\delta)(1+\tau)]k - (1+\tau)k' + T.$$

Hence, the typical household's problem can be written recursively as

$$\begin{aligned} V(k,\bar{k}) &= \max_{c,k'} \Big\{ u(c) + \beta V(k',\bar{k}') \Big\} \\ s.t. \;\; c &= w + [r + (1-\delta)(1+\tau)]k - (1+\tau)k' + T, && (1) \\ \bar{k}' &= H(\bar{k}), && (2) \\ w &= w(\bar{k}) = F_2(\bar{k},1), && (3) \\ r &= r(\bar{k}) = F_1(\bar{k},1), && (4) \\ T &= T(\bar{k}) = \tau[H(\bar{k}) - (1-\delta)\bar{k}], && (5) \end{aligned}$$

where $k$ is the individual capital, and $\bar{k}$ is the aggregate capital. Moreover, (1) is the household's budget constraint, (2) is the aggregate law of motion of capital, (3) and (4) follow directly from market clearing, and (5) is the government revenue as a function of the aggregate state.

The definition of a RCE is standard, and can be found in the lecture notes. In the definition of RCE, the most important part is to clarify that consistency requires $g(\bar{k},\bar{k}) = H(\bar{k})$, where $H$ was defined above, and $g$ is the typical household's policy function.

b) The Euler equation for the typical household is given by

$$u'(c)(1+\tau) = \beta[F_1(\bar{k}',1) + (1-\delta)(1+\tau)]\,u'(c').$$

But the objective here was to express everything as a function of the aggregate capital stock only. To that end, notice that we can use the budget constraint to write consumption in a more useful form. First impose consistency (i.e., $k = \bar{k}$) to obtain

$$c = w + [r + (1-\delta)(1+\tau)]\bar{k} - (1+\tau)\bar{k}' + T.$$

Next, replace the prices and the total revenue $T$ with the terms that include the aggregate capital stock (i.e., $r = F_1(\bar{k}, 1)$, $w = F_2(\bar{k}, 1)$, and $T = \tau[\bar{k}' - (1-\delta)\bar{k}]$). We obtain:

$$\begin{aligned} c &= F_2(\bar{k},1) + [F_1(\bar{k},1) + (1-\delta)(1+\tau)]\bar{k} - (1+\tau)\bar{k}' + \tau[\bar{k}' - (1-\delta)\bar{k}] = \\ &= F_2(\bar{k},1) + F_1(\bar{k},1)\bar{k} + (1-\delta)\bar{k} - \bar{k}'. \end{aligned}$$

Finally, exploiting Euler's Theorem and the definition of $f(\bar{k})$, we find that

$$c = f(\bar{k}) - \bar{k}'.$$

Using this expression back into the Euler equation, and noticing that $f'(\bar{k}) = F_1(\bar{k}, 1) + 1 - \delta$, we obtain the second-order difference equation, which describes the law of motion of aggregate capital:

$$u'(f(\bar{k}) - \bar{k}')(1+\tau) = \beta[f'(\bar{k}) + \tau(1-\delta)]\, u'(f(\bar{k}') - \bar{k}'').$$

c) In the steady-state equilibrium $c = c'$ and $\bar{k} = \bar{k}' = \bar{k}''$. Imposing this condition on the Euler equation, it is easy to show that the steady state capital stock is given by:

$$\bar{k}^*(\tau) \equiv \left\{ \bar{k} : f'(\bar{k}) = \frac{1 + \tau[1 - \beta(1-\delta)]}{\beta} \right\}. \tag{6}$$

d) With $F(\bar{k}, n) = A\bar{k}^a n^{1-a}$, we have $f(\bar{k}) = A\bar{k}^a + (1-\delta)\bar{k}$ and $f'(\bar{k}) = aA\bar{k}^{a-1} + 1 - \delta$. Using these functional forms in (6), we get

$$\bar{k}^*(\tau) = \left[ \frac{aA\beta}{(1+\tau)[1-\beta(1-\delta)]} \right]^{\frac{1}{1-a}}. \tag{7}$$

e) As I was trying to point out in the hint, your first guess does not need to be sophisticated, it needs to be easy! And there is no easier way to start than by assuming that $V_0(k, \bar{k}) = 0$. Yes, that's a terribly wrong guess, but it's the one that will generate the easiest math. It is also the initial guess we assumed in the related problem set. (See the second problem in Problem Set 1.) Given our initial guess, we now have:

$$V_1(k, \bar{k}) = \max_{k'} \left\{ ln\left((1-a)A\bar{k}^a + aA\bar{k}^{a-1}k - k'\right) + 0 \right\},$$

and obviously the optimal $k'$ is also 0. (Because the logarithmic function is strictly increasing, and the term $k'$ appears only once in that term and enters multiplied by a minus.) Imposing this optimal $k'$ in the last expression, we conclude that

$$V_1(k, \bar{k}) = ln\left((1-a)A\bar{k}^a + aA\bar{k}^{a-1}k\right).$$

This is the part where we need to a little algebra and re-write our $V_1$ as

$$V_1(k,\bar{k}) = ln(aA) + ln\left(k + \frac{1-a}{a}\,\bar{k}\right) + (a-1)\,ln\left(\bar{k}\right). \tag{8}$$

Of course, we should not expect that our value function will converge after just one iteration! But, quite interestingly (and as suggested in the hint), after just one iteration the VF has already taken the correct form:

- It starts with *some* constant

- Then it has a term that involves both the aggregate *and* the individual state, and takes the form $ln\left(k + \bar{k}\text{ “}\textit{times something}\text{”}\right)$

- Third, it has a term that depends only on $ln(\bar{k})$ and is multiplied by some other constant

Again, I did not want you to find those arbitrary constants; I just wanted you to guess the correct form, and it is now transparent that the correct VF is of the form:[1]

$$V(k,\bar{k}) = \gamma_0 + \gamma_1\,ln\left(k + \gamma_2\,\bar{k}\right) + \gamma_3\,ln\left(\bar{k}\right).$$

f) We simply need to impose our guess from part (e) into the Bellman equation, as suggested in the hint, and take FOC with respect to $k'$. Under our educated guess in part (e), the Bellman equation for the household who takes prices and the aggregate law of motion of capital as given, becomes:

$$V(k,\bar{k}) = \max_{k'}\left\{ln\left((1-a)A\bar{k}^a + aA\bar{k}^{a-1}k - k'\right) + \beta\left[\gamma_0 + \gamma_1\,ln\left(k' + \gamma_2\,\bar{k}'\right) + \gamma_3\,ln\left(\bar{k}'\right)\right]\right\}.$$

Taking the FOC with respect to the households *individual* capital choice for tomorrow, $k'$, we obtain:

$$\frac{1}{aA\bar{k}^{a-1}k + (1-a)A\bar{k}^a - k'} = \frac{\beta\gamma_1}{k' + \gamma_2\bar{k}'}.$$

Before we solve for $k'$, notice that a proper policy function must give us the optimal choice for $k'$ as a function of the two states, $k,\bar{k}$, only. In the above expression, we also see $\bar{k}'$, which is problematic. But in the hint I encouraged you to guess that $\bar{k}' = s\,A\bar{k}^a$.

[1] To be completely fair, after just one iteration, it seems like the term $ln\left(k + \gamma_2\,\bar{k}\right)$ is not multiplied by anything. (See equation 8.) In other words, it seems like $\gamma_1 = 1$. If you had tried one more iteration, you would have clearly seen that this term indeed has a multiplier $\gamma_1 \neq 1$. But I did not want you to spend too much time doing numerous iterations, especially since these iterations get harder and harder. Generally speaking, assuming that the term $ln\left(k + \bar{k}\text{ “}\textit{times something}\text{”}\right)$ has a multiplier $\gamma_1$ is a good idea, since the worst thing that can happen is that you will end up finding that $\gamma_1 = 1$! On the other hand, assuming that this term does *not* have a multiplier is not harmless. Indeed, in this case, $\gamma_1$ is not equal to 1. But, I decided to give full credit to students who guessed that the form of the Bellman equation was $V(k,\bar{k}) = \gamma_0 + ln\left(k + \gamma_1\,\bar{k}\right) + \gamma_2\,ln\left(\bar{k}\right)$.

Imposing this (educated) guess, and after some algebra, we conclude that the policy function of the representative household takes the form:

$$k' = g(k, \bar{k}) = \frac{\beta\gamma_1(1-a) - \gamma_2 s}{1+\beta\gamma_1} A\bar{k}^a + \frac{\beta\gamma_1}{1+\beta\gamma_1} aA\bar{k}^{a-1}k.$$

This is our guess for the policy function that depends only on the unknown $\gamma$ parameters (and the $s$ from our last conjecture), and the two state variables, $k, \bar{k}$. Solving for these unknown parameters is not hard, but this is not what I was asking you in this question.

**Question 2**

a) The typical buyer's enters the CM with some money, $m$, and some debt, say $b$. Her value function is given by

$$W(m,b) = \max_{X,H,\hat{m}} \Big\{ X - H + \beta V(\hat{m}) \Big\},$$
$$s.t.\ \ X + \phi\hat{m} = \phi m + H - b + T,$$

where $T$ is the lump-sum monetary transfer to the buyer by the monetary authority. Using the standard procedure described in class, we will find that

$$W(m,b) = \Lambda + \phi m - b,$$

where $\Lambda$ just summarizes a number of terms that are not related to the state variables.

As we know from class, the sellers' VF are also linear, even though here I did not ask you to explicitly prove that.

b) Bargaining in a type-1 meeting is identical to the one we saw in class. The buyer wishes to purchase the first-best amount $q^*$, but she may be constrained by her cash holdings. Thus, the bargaining solution is as follows: $q_1 = \min\{\phi m, q^*\}$ and $d_1 = \min\{m, q^*/\phi\}$.

In a type-2 meeting things are very similar, except that the buyer can also purchase the day good using some credit, up to the amount $C < q*$. Thus, it turns out that there exist a critical level of money, call it $m_2^*$, such that buyers who carry that amount of money or more, can get $q^*$ in the type-2 meeting. What is that amount? Clearly, it is

$$m_2^* \equiv \frac{q^* - C}{\phi}.$$

Then, $q_2 = \min\{C + \phi m, q^*\}$ and $d_2 = \min\{m, m_2^*\}$.

c) As I explained, there is no need to formally derive the objective because it is quite intuitive (assuming one has understood the environment well). The one thing to notice (highlighted in the footnote), is that the objective function will behave differently around the point $m_2^* = (q^* - C)/\phi'$.[2] Intuitively, if the buyer chooses an $m'$ higher than $m_2^*$, she can afford $q^*$ in the event of meeting a type-2 seller (but not in the event of meeting a type-1 seller). If the buyer chooses $m' < m_2^*$, then she cannot afford $q^*$ even if she met a type-2 seller.[3] Therefore, the objective function can be summarized as follows.

[2] Clearly, $m_2^*$ is defined in an identical fashion as above, except it contains $\phi'$ in the denominator, because the objective function summarizes the optimal money holdings *for the next period*.

[3] As we discussed in class, with $i > 0$, the buyer will never choose $m' > q^*/\phi'$. Another way of saying this is that in the type-1 meeting the buyer will always spend all her money, or that the bargaining solution will be in the "binding branch", i.e., $d_1 = m$ and $q_1 = \phi m$.

- If $m' \geq m_2^*$, then

$$J(m') = (-\phi + \beta\phi')m' + \beta(1-\sigma)[u(\phi' m') - \phi' m'] + \beta\sigma[u(q^*) - (q^* - C)],$$

- If $m' < m_2^*$, then

$$J(m') = (-\phi + \beta\phi')m' + \beta(1-\sigma)[u(\phi' m') - \phi' m'] + \beta\sigma[u(\phi' m' + C) - \phi' m'],$$

d) The next step is to obtain the FOC and evaluate it in equilibrium.

For $m' \geq m_2^*$, the FOC yields:

$$\phi = \beta\phi' \Big\{ 1 + (1-\sigma)[u'(\phi' m') - 1] \Big\}.$$

For $m' < m_2^*$, the FOC yields:

$$\phi = \beta\phi' \Big\{ 1 + (1-\sigma)[u'(\phi' m') - 1] + \sigma[u'(\phi' m' + C) - 1] \Big\}.$$

In the steady state equilibrium, we have $z = \phi M = \phi' M'$. Moreover, using the Fisher equation, we can replace the term $(1 + \mu - \beta)/\beta$ with the nominal interest rate on an illiquid bond, $i$. Hence, the two earlier FOCs will become:

$$i = (1-\sigma)[u'(z) - 1], \tag{9}$$
$$i = (1-\sigma)[u'(z) - 1] + \sigma[u'(C + z) - 1], \tag{10}$$

which give us the equilibrium real balances, $z$, as a function of $i$. Only one question remains: for which $i$'s is (9) the relevant equilibrium condition, and for which ones should we refer to (10)? (This is precisely the question I directed you to ask in the Hint.)

Condition (10) contains an extra term, which is the marginal utility of an additional dollar to a buyer who meets with a type-2 seller. That term is there (and is positive) precisely because that buyer did not carry too many dollars, so any additional dollar has a high valuation because it is helpful in BOTH types of meetings. (For the sake of discussion, let's call this the "scarce money" case). Of course, as the buyer carries more money, that term eventually disappears, because beyond a certain level of $z$, the buyer who meets a type-2 seller will be able to afford $q^*$. For that buyer the marginal benefit of an extra dollar, in the event of meeting a type-2 seller, is zero. (For the sake of discussion, let's call this the "plentiful money" case). What is the level of $i$ that separates the space between the scarce and plentiful cases? Well, of course, the $i$ for which $z + C = q^*$, or, equivalently,

$$\bar{\iota} \equiv (1-\sigma)[u'(q^* - C) - 1].$$

To summarize,

- For any $i \leq \bar{\iota}$, equilibrium $z$ is given by (9);
- For any $i > \bar{\iota}$, equilibrium $z$ is given by (10).

e) Given our work in part (d) this is very easy. Of course, $q_1 = z$: with TIOLI offers by the buyer, the amount she will purchase in a type-1 meeting is exactly equal to her real balances.

How about $q_2$? If the interest rate is relatively low, the buyer will carry enough balances so that these balances together with the credit will get her $q^*$. If the interest rate is high, the buyer will give up all her $z$ and still not afford $q^*$. Formally,

- If $i \leq \bar{\iota}$, $q_2 = q^*$;
- If $i > \bar{\iota}$, $q_2 = z + C < q^*$.

f) As I explained in the question, the welfare function here is

$$\mathcal{W} = \sigma[u(q_2) - q_2] + (1-\sigma)[u(q_1) - q_1],$$

where $q_1, q_2$ were determined in the previous parts. Also, we assume a quadratic utility. It is easy to verify that with these specific preferences, we have $q^* = \gamma$ and $\bar{\iota} = (1-\sigma)C$. Hence, equilibrium is as follows:

- If $i \leq \bar{\iota}$, $q_1 = z = \gamma - \frac{i}{1-\sigma}$ and $q_2 = q^* = \gamma$;
- If $i > \bar{\iota}$, $q_1 = z = \gamma - i - \sigma C$ and $q_2 = \gamma - i + (1-\sigma)C$.

Now, recall that the question concerns the derivative of $\mathcal{W}$ with respect to $C$. If $i \leq \bar{\iota}$, then the equilibrium is irrelevant from $C$, so this is not an interesting case. So let's focus on the "scarce" case where interest rates are relatively high. Here,

$$\mathcal{W} = \sigma[u(\gamma - i + (1-\sigma)C) - (\gamma - i + (1-\sigma)C)] + (1-\sigma)[u(\gamma - i - \sigma C) - (\gamma - i - \sigma C)].$$

Exploiting the quadratic utility form one more time, and differentiating with respect to $C$ yields (after some algebra):

$$\frac{\partial \mathcal{W}}{\partial C} = -\sigma(1-\sigma)C,$$

which is negative for any parameter values. The intuition is as follows: a higher credit limit, $C$, sounds like a good idea because it implies that *ex post*, buyers who meet type-2 sellers will be able to get a very high $q_2$. But *ex ante* it discourages agents from carrying money. It turns out that with this functional form, the second force is so powerful that welfare ends up being decreasing in $C$ for any parameter values.